
Towards Preference-Aligned 3D Quality Assessment

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Abstract

3D content creation is rapidly expanding across applications such as AR/VR, robotics, and simulation, yet evaluating the perceptual quality of 3D assets remains challenging. Existing 3D quality assessment (3D-QA) benchmarks mainly rely on synthetic distortions, which fail to capture naturally occurring artifacts in real-world and generative 3D data. In this work, we introduce 3D-PAQA, the first large-scale preference-aligned 3D-QA dataset constructed using multimodal large language models (MLLMs) guided by human-labeled exemplar anchors. By leveraging exemplar-anchored prompting and relative ranking, our approach injects human preference signals into the MLLM’s context and yields stable, human-aligned quality annotations for over 260K 3D assets across six perceptual criteria. We further train a lightweight 3D-Evaluator based on Point Transformer-v3 that predicts perceptual quality directly from point-cloud features. Remarkably, the evaluator surpasses its teacher MLLM in correlation with human judgments, indicating that MLLM-derived supervision can be distilled into a compact, robust metric. Beyond benchmarking, 3D-PAQA and its evaluator pave the way for scalable, human-aligned 3D assessment applicable to dataset curation and generative model evaluation. We further demonstrate the evaluator as an efficient, human-aligned reward inside an agentic generation loop, where its decomposed criterion-wise feedback steers an LLM agent to iteratively refine generated 3D assets.

1 Introduction

3D content creation is rapidly gaining attention across applications such as AR/VR, robotics, and simulation. With the emergence of large-scale repositories like Objaverse and ShapeNet [2, 1], 3D assets have become widely accessible, fueling the recent surge in 3D generative modeling. However, assessing the quality of these assets remains a significant challenge. Ensuring high-quality meshes is essential for the usability of downstream 3D applications, yet existing benchmarks often lack meaningful quality indicators.

Most existing 3D quality assessment (3D-QA) approaches are focusing on synthetic distortions [10, 12, 36], where artificial degradations (*e.g.*, *gaussian noise*, *downsampling*) are introduced to generate distorted samples from pristine references. While useful for controlled experiments, such distortions poorly reflect the artifacts commonly found in real-world 3D assets (See Figure 1.). Consequently, metrics trained on distortion-based datasets often misalign with human perception, highlighting the need for more realistic and subjective evaluation.

The same challenge arises in 3D generative modeling. As shape generation or text-to-3D models advance, evaluating their outputs remains difficult due to the absence of clear ground truth. Although user studies are commonly employed, they are costly, inconsistent, and unscaleable. Moreover, generative model training itself requires careful quality control: low-quality assets can degrade performance, while preference-aligned QA can enable better dataset curation. Thus, scalable human-aligned datasets are essential for both evaluating and curating 3D content.

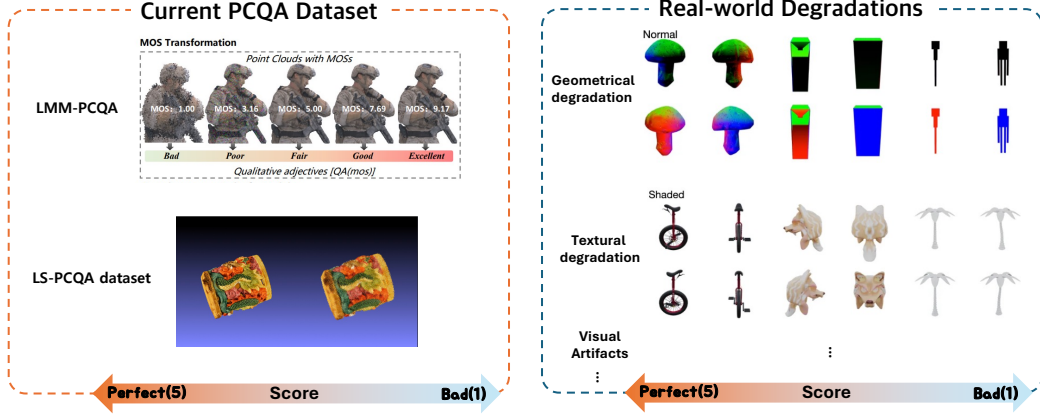


Figure 1: **Dataset degradation comparisons.** Existing 3D-QA datasets rely on synthetic distortions (e.g., geometrical or textural degradation), which fail to represent real-world artifacts (The figure is copied from [15, 10]). In contrast, 3D-PAQA captures naturally occurring degradations and aligns quality assessment with human perception through MLLM-based preference annotations.

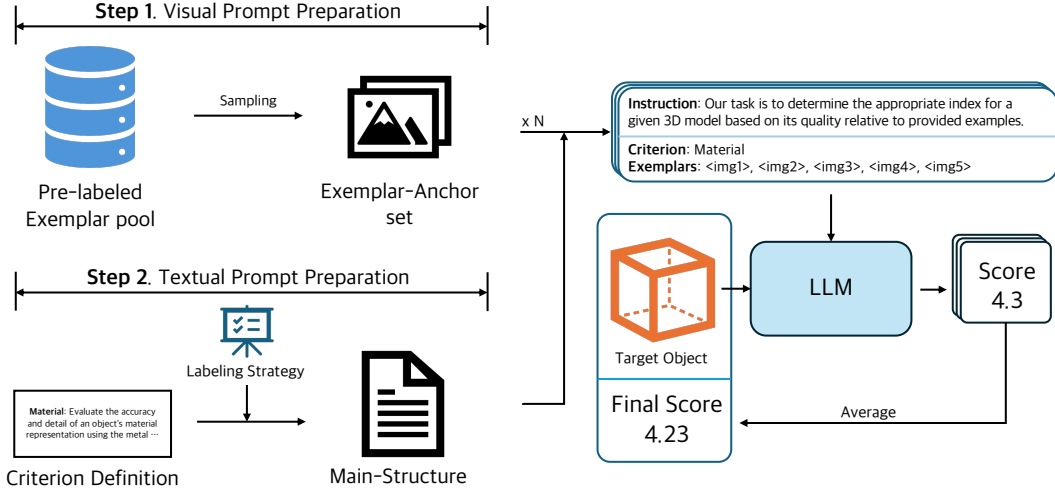


Figure 2: **Overview of the 3D-PAQA dataset construction pipeline.** Multi-view renderings of each object are evaluated through exemplar-anchored visual prompts and Labeling Strategy by an MLLM, yielding stable, preference-aligned annotations across six perceptual criteria. The exemplar anchor set is sampled N times, resulting in N quality scores predicted by the LLM for each criterion. These scores are then averaged to obtain the final annotation for the target object.

To address these challenges, recent studies[5, 8, 22, 32] have explored automated 3D evaluation frameworks leveraging large models to approximate human perception. These can be broadly categorized into MLLM-based and non-MLLM-based approaches. MLLM-based methods exploit large vision-language models such as GPT-4V to assess 3D quality via pairwise A/B testing or instruction-based comparisons, demonstrating strong alignment with human judgment [32, 5, 22]. However, these methods require extensive comparisons and incur high computational costs, while still producing relative rather than absolute quality scores. In contrast, non-MLLM methods such as Objaverse++ [8] train lightweight curators on small human-labeled datasets ($\approx 10k$ samples), but their limited scale restricts generalizability. Our work combines the strengths of both paradigms—leveraging the human-alignment capability of MLLMs to construct a scalable preference-aligned dataset, which also enables the training of an efficient 3D evaluator.

In this paper, we propose 3D Preference-Aligned Quality Assessment (3D-PAQA), a new framework for scalable and human-aligned 3D quality evaluation. Our goal is to construct a dataset and evaluation pipeline that capture subjective human preferences without relying on costly user studies or distortion-

based assumptions. To this end, we leverage multimodal large language models (MLLMs) to annotate large-scale 3D assets through a structured prompting scheme. Specifically, we design a two-stage system consisting of (1) exemplar-anchored visual prompting, where human-labeled reference samples inject preference anchors into the MLLM’s context, and (2) a relative ranking mechanism that stabilizes quality judgments across six perceptual dimensions (See Figure 2). This process yields a large-scale, preference-consistent 3D-QA dataset covering over 260K assets and enables the training of a lightweight 3D-Evaluator that distills MLLM annotations into an efficient point-based quality predictor. Compared to prior frameworks that depend on pairwise A/B testing or limited user labels, 3D-PAQA achieves improved scalability, stability, and human alignment, establishing a practical foundation for both dataset curation and generative model evaluation.

Overall, our work demonstrates that it is possible to construct a scalable, human-aligned 3D-QA dataset by leveraging MLLMs. At the same time, we formally introduce the 3D-PAQA task, establishing a benchmark and methodology for preference-aligned 3D quality assessment. Specifically, we make the following contributions:

- We present **the first large-scale human-aligned 3D quality dataset** with annotations for **260k** Objaverse assets across six criteria. This explicitly bridges the gap between subjective human aesthetic preferences and objective automatic QA in 3D.
- To generate large-scale annotations, we design a **simple in-context prompting scheme** that embeds exemplar-anchor sets into MLLM prompts, enhancing label stability and consistency without costly A/B testing.
- We show that a **light-weight 3D evaluator** trained on our dataset can efficiently predict preference-aligned quality scores, demonstrating superior performance compared to a 72B-parameter MLLM (Qwen2-VL 72B) on a human-labeled benchmark.
- We demonstrate the evaluator as a **verifier for agentic 3D generation**: its criterion-wise scores act as an efficient, human-aligned reward that steers an LLM agent to iteratively refine generated assets, raising the initially weakest criterion at negligible cost.

2 Related work

3D Quality Assessment and datasets. 3D quality assessment (3D-QA) aims to evaluate the perceptual quality of distorted 3D content. Most existing methods focus on synthetic distortions introduced for controlled degradations such as compression, transmission, or rendering. Benchmarks like WPC, LS-PCQA, and SJTU-PCQA [10, 12, 36] adopt this design, where pristine references are augmented with distortions (e.g., downsampling, Gaussian noise, compression) and scored by subjective ratings. To mitigate the scarcity of dataset scale, various methods have been proposed [13, 18, 19, 37, 39]: DiSPA disentangles perceptual factors via mutual information minimization, CoPA leverages contrastive pre-training on unlabeled point clouds, and IT-PCQA transfers perceptual knowledge from 2D IQA through domain adaptation. However, synthetic distortions poorly reflect real-world artifacts (See Figure 1.)—such as non-manifold geometry or unrealistic proportions—and the limited diversity of reference objects restricts generalization. This motivates moving beyond synthetic benchmarks. In this work, we release the first large-scale 3D-QA dataset capturing naturally occurring artifacts in generated 3D assets. Comprising over 260k unique samples, our dataset enables realistic and scalable quality assessment, bridging subjective human preferences with automatic evaluation across diverse and unseen 3D content.

3D Generative models. 3D generative models have advanced rapidly, fueled by large-scale datasets such as Objaverse [2]. Existing approaches fall into three categories: optimization-based methods [7, 11, 16, 25], multi-view diffusion-based methods [4, 9, 20, 21, 31], and large reconstruction models [23, 26, 34, 35, 38]. Recent works such as TRELLIS [34] and CLAY [38] indicate that 3D models are beginning to approach human-level quality in generating realistic content. Despite this progress, evaluation and curation remain bottlenecks. Current assessments of generated assets often rely on statistical metrics such as 3D-IoU, FID or F-score, which fail to reflect human preferences and thus require costly user studies [31, 38]. At the same time, the quality of training data strongly influences generative performance. Since Objaverse is web-crawled, its assets vary widely in quality, leading researchers to adopt ad-hoc curation strategies ranging from manual filtering to CLIP-based heuristics. These approaches are either labor-intensive, computationally costly, or indirect, underscoring the need for scalable, preference-aligned 3D quality assessment.

MLLMs as human-aligned evaluators. The emergence of CLIP marked a milestone in multimodal learning by aligning vision and language within a shared space, and GPT-4V further demonstrated that MLLMs can interpret visual inputs in ways closely aligned with human perception. Building on this capability, recent works in IQA [27–30] showed that MLLMs can function as human-aligned quality evaluators, bridging the gap between traditional distortion-based metrics and subjective human judgment. This trend has recently extended to 3D [5, 22, 14, 32], with approaches applying MLLMs to assess generative models. While these methods highlight the potential of preference-aligned 3D evaluation, they remain limited to comparisons between models rather than per-content scoring. Moreover, most rely on A/B testing aggregated into Elo ratings [3], which require a large number of pairwise comparisons, making them computationally costly and inherently heuristic. This reliance on relative scoring underscores a fundamental limitation of current MLLM-based evaluators in 3D QA.

3 3D-PAQA

The goal of this work is to achieve scalable and human-aligned 3D quality assessment by constructing a large-scale preference-aligned dataset and training an efficient 3D evaluator. To build 3D-PAQA, we leverage MLLMs guided by a stratified pool of human-labeled exemplar anchors that inject human preference into the prompting process. Conditioned on these anchors, the MLLM predicts perceptual scores for target samples, producing stable annotations used for evaluator training.

A key challenge in preference-based quality assessment lies in its inherent *subjectivity*, often making MLLM outputs unstable and inaccurate. To address this, we introduce two core components: **Exemplar-Anchors** and a **Relative Ranking strategy**, which jointly enforce consistency in MLLM judgments and yield more reliable preference-aligned annotations.

3.1 Task Definition: Preference-aligned 3DQA

3D quality assessment (3D-QA) traditionally aims to measure the perceptual fidelity of 3D assets, often with respect to synthetic distortions such as *downsampling*, *Gaussian noise*, or *compression artifacts* [10, 12, 13]. While such distortion-based benchmarks provide controlled testbeds, they fail to capture the naturally occurring artifacts—including non-manifold geometry, unrealistic proportions, or missing surfaces—that are prevalent in real-world 3D content and especially in assets generated by modern 3D generative models.

To bridge this gap, we define the task of **3D Preference-Aligned Quality Assessment (3D-PAQA)**. Unlike *distortion-centric* QA, 3D-PAQA explicitly aligns quality evaluation with human perceptual preference. Formally, given a 3D asset X , the goal is to estimate a quality score vector

$$Q(X) = [q_1, q_2, \dots, q_k] \quad (1)$$

across k perceptual criteria (e.g., geometry plausibility, texture fidelity, material consistency, artifact absence), where each q_i reflects human-aligned judgment.

The core distinction is that 3D-PAQA does not assume the availability of pristine references or artificial distortions. Instead, it targets absolute quality estimation of *arbitrary* 3D content—both curated datasets and generative outputs—anchored to subjective human preferences. This makes 3D-PAQA directly applicable to benchmarking generative models, curating large-scale repositories (e.g., Objaverse[2]), and improving training pipelines where data quality critically impacts downstream performance.

3.2 Dataset Construction with MLLM

To build a large-scale preference-aligned dataset, we leverage multimodal large language models (MLLMs) as scalable annotators. However, directly prompting MLLMs to score 3D assets often produces unstable and inconsistent results, due to the inherently subjective nature of preference-based assessment.

To address this challenge, we design a two-stage construction pipeline consisting of visual prompting with exemplar-anchors and a preference-assessing strategy (See Figure 3.).

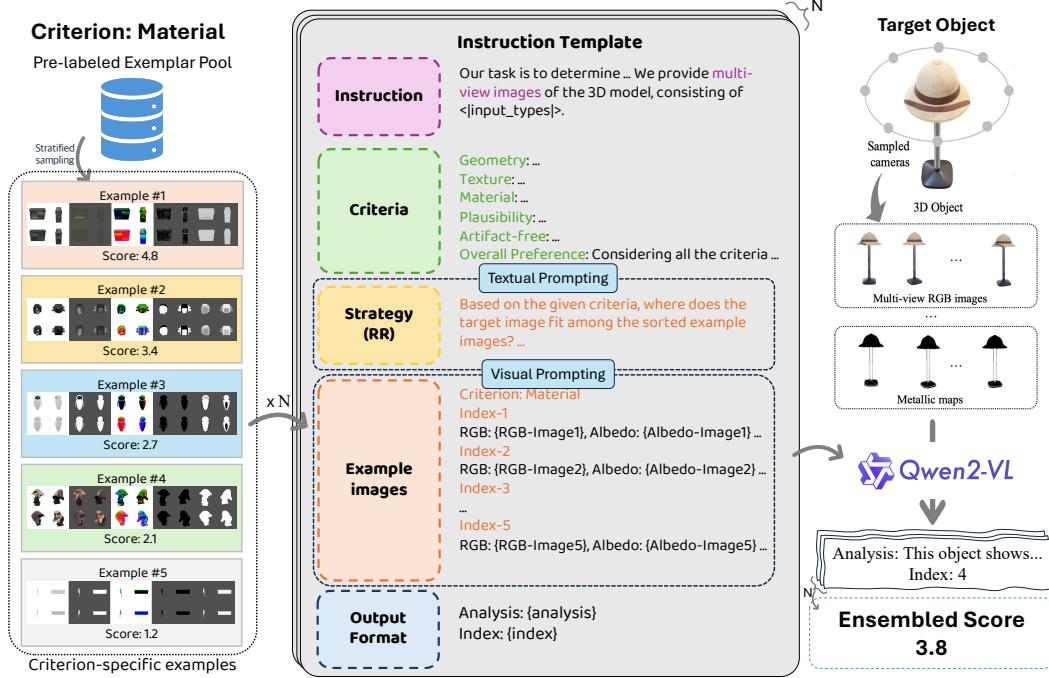


Figure 3: **Overview of the 3D-PAQA annotation process.** Our pipeline leverages multimodal large language models (MLLMs) to generate stable preference-aligned annotations. In the visual prompting stage, a set of stratified exemplar anchors—each representing a distinct quality level—are sampled from a human-labeled pool and presented alongside the target 3D asset. In the textual prompting stage, the MLLM receives criterion-specific instructions (e.g., geometry, texture, material) and predicts the relative rank of the target asset among exemplars. Repeating this process with multiple anchor sets and averaging the results yields consistent, human-aligned quality scores across six perceptual criteria.

Visual Prompting: Exemplar-Anchoring injection. Each target 3D asset is rendered into a set of multi-view images, which are fed to the MLLM along with carefully designed visual and textual instructions (See Figure 3-a). To stabilize outputs, we inject a set of *human-labeled exemplar anchors* into the prompt.

Concretely, we first prepare 100 pre-labeled reference objects spanning 10 categories (10 objects per category), and group them into five quality levels (high to low, 20 objects each). For each evaluation, we sample one object from each group, yielding a set of five anchors that cover the entire quality spectrum. These anchors are inserted alongside the query asset, providing explicit *reference points* for the MLLM to ground its prediction. To reduce anchor-specific bias, we repeat this evaluation five times with independently sampled anchors and average the resulting scores. This exemplar-anchoring scheme ensures that preference signals are explicitly embedded in the prompt and that final annotations remain consistent and robust across variations in anchor choice.

Textual Prompting: Absolute vs. Relative Labeling. To generate preference-aligned annotations at scale, we explored two strategies for scoring 3D assets (See Figure 4). The first is **Absolute Scoring (AS)**, where the MLLM directly predicts an absolute score (e.g., 1–5) for each criterion. While it is straightforward, AS often suffers from instability due to the subjective nature of quality assessment.

To obtain more reliable labels, our main dataset construction relies on a **Relative Ranking (RR)** strategy. Here, the MLLM compares the target asset against exemplar anchors and predicts its relative position within the set, from which the prediction score is derived by averaging the scores of the nearest anchors. This relative grounding induces MLLM to be more grounded on the given exemplar anchors, making the outputs more stable and consistent, which is crucial for constructing reliable annotations at scale.

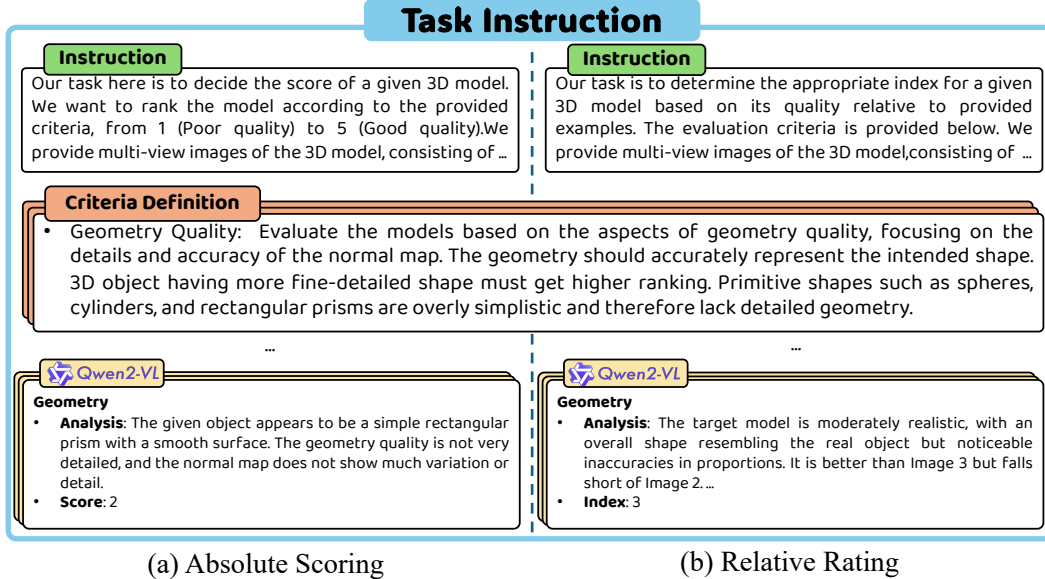


Figure 4: **Comparison of annotation strategies in 3D-PAQA.** We compare two methods for generating preference-aligned annotations using MLLMs. (a) **Absolute Scoring (AS):** the MLLM directly assigns an absolute quality score (1–5) to the target asset for each criterion based on its overall analysis. (b) **Relative Ranking (RR):** the MLLM ranks the target asset among exemplar anchors of varying quality, then derives the final annotation by averaging the scores of the nearest anchors. This relative scheme reduces subjectivity and produces more stable, consistent annotations across perceptual dimensions such as geometry, texture, and material.

3.3 3D Evaluator Training

Having established 3D-PAQA as a scalable and human-aligned dataset, we further validate its effectiveness by training a lightweight 3D evaluator on our annotations. Each 3D asset is sampled into a normalized point set and encoded using a Point Transformer-v3 backbone [33], followed by a small regression head that jointly predicts k preference-aligned quality scores across all perceptual criteria. The model is optimized using a Huber regression loss [6] against our 3D-PAQA annotations, without introducing any additional architectural complexity.

This experiment demonstrates that 3D-PAQA not only provides large-scale, human-aligned supervision but also enables the training of practical and efficient downstream evaluators. The resulting model achieves strong correlation with human-labeled benchmarks while being orders of magnitude smaller and faster than MLLM-based evaluators, confirming that our dataset serves as a reliable foundation for scalable 3D quality assessment (Section 4.3).

4 Experiments

4.1 Evaluation protocol

We report Spearman (SROCC), Pearson (PLCC), and Kendall’s τ correlations (KROCC) with human-labeled subsets that were not included in evaluator training. To establish a benchmark, we conducted a user study on an Objaverse subset covering all six criteria, collecting 12k human responses. Baselines include zero-shot MLLMs (Qwen2-VL-72B, Qwen2-VL-7B) and curator-based methods (Objaverse++) for dataset analysis.

4.2 Dataset Analysis

Experimental Details. We implement an annotation pipeline that queries MLLM (Qwen2-VL-72B-int) with exemplar-anchored prompts [24]. For each object, we load its multi-view renderings and PBR maps (RGB, normal, roughness, metallic, albedo), and augment the prompt with five

Method		Preference		Plausibility		Artifacts		Geometry		Texture		Material		Average	
# of param	Strategy	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
7B	AS	0.416	0.488	0.608	0.566	0.366	0.249	0.583	0.552	0.797	0.711	0.540	0.568	0.552	0.522
	RR	0.372	0.245	0.401	0.275	0.276	0.045	0.528	0.500	0.591	0.595	0.183	0.163	0.392	0.354
72B	no-exp	0.582	0.605	0.586	0.578	0.413	<u>0.463</u>	<u>0.620</u>	<u>0.658</u>	0.794	0.772	<u>0.629</u>	<u>0.566</u>	0.604	0.607
	AS	<u>0.648</u>	0.690	<u>0.638</u>	<u>0.614</u>	<u>0.431</u>	0.431	0.599	0.611	0.871	0.839	0.639	0.629	0.638	0.636
	RR	0.685	<u>0.671</u>	0.661	0.663	0.495	0.544	0.567	0.534	<u>0.851</u>	<u>0.724</u>	0.530	0.514	<u>0.631</u>	<u>0.608</u>

Table 1: **Comparison of MLLM-annotated datasets.** We report PLCC and SROCC correlations across six perceptual criteria with human-labeled subsets, along with their averages.(no-exp: no exemplar-anchor, AS: Absolute Scoring strategy, RR: Relative Ranking strategy) Bold indicates best results, underline indicates second-best.

Method	Preference		Plausibility		Artifacts		Geometry		Texture		Material	
	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
72B-RR	0.669	0.657	<i>0.646</i>	<i>0.649</i>	<i>0.474</i>	<i>0.520</i>	<i>0.564</i>	<i>0.544</i>	<i>0.848</i>	<i>0.723</i>	<i>0.519</i>	<i>0.497</i>
Objaverse++ [8]	0.505	0.514	–	–	–	–	–	–	–	–	–	–
3D Evaluator	0.773	0.777	<i>0.659</i>	<i>0.662</i>	<i>0.502</i>	<i>0.567</i>	<i>0.653</i>	<i>0.617</i>	<i>0.903</i>	<i>0.806</i>	<i>0.646</i>	<i>0.659</i>

Table 2: **Comparison between 3D-PAQA dataset, Objaverse++ and our 3D Evaluator.** Bold indicates best result, blue italics denote sub-criteria metrics.

stratified exemplar anchors spanning the quality spectrum. To mitigate anchor-specific bias, we repeat the evaluation with five independent anchor sets and average the outputs to obtain robust annotations. This procedure yields preference-aligned quality labels for 260k Objaverse assets across six perceptual criteria (geometry, texture, material, 3D plausibility, artifacts, and overall preference). Additional implementation details are provided in the Appendix.

Scale and Coverage. We build 3D-PAQA upon the gobjaverse dataset, which provides 280K categorized and publicly available multi-view renderings and PBR assets[17]. From the total 265K usable samples, we leverage the provided category taxonomy spanning 10 semantic domains: *buildings and outdoor scenes, daily-used items, transportation, poor-quality artifacts, human-shaped assets, electronics, animals, furniture, plants, and food*. For each category, 10 representative samples are selected to form a 100-object exemplar-anchor pool, which serves as the human reference set during annotation. Another 100 samples (10 per category) are separately drawn to construct an evaluation set for validating annotation consistency. The remaining data are annotated through our MLLM pipeline under five configurations — 7B-AS, 7B-RR, 72B-no-exp, 72B-AS, and 72B-RR — corresponding to different model capacities and prompting strategies. This design provides semantically diverse and category-balanced annotations, supporting comprehensive and domain-agnostic evaluation of 3D quality.

Human-Alignment ability. We evaluate five annotation configurations (7B-AS, 7B-RR, 72B-no-exp, 72B-AS, 72B-RR) using the Qwen2-VL family. The results are summarized in Table 1. We observe that the 7B model exhibits a significant performance drop when exemplar-anchoring prompts are introduced, particularly under the relative-ranking (RR) strategy. This indicates that smaller models struggle to process multi-reference reasoning within complex prompts, which require higher model capacity. In contrast, exemplar anchoring substantially improves correlation scores for the 72B model, demonstrating its ability to interpret visual anchors effectively.

Moreover, the RR strategy consistently outperforms AS for *holistic* criteria such as preference, plausibility, and artifacts, while AS excels in *component-level* attributes like geometry, texture, and material. Interestingly, the no-exp configuration yields competitive scores in geometry and material, suggesting that these most structural attributes can be objectively assessed without exemplar guidance. For all subsequent experiments, we adopt the 72B-RR annotations as the primary training supervision, given their strongest human-aligned preference correlation.

Our 3D-PAQA has a correlation with object’s complexity. We further analyze whether the annotated preference scores correlate with the intrinsic complexity of the 3D assets. Specifically,

72B-RR	# of assets	Avg. of Vertex #	Avg. of Face #
$Score = 5$	2762	198737	207918
$4 \leq Score < 5$	35227	166821	188674
$3 \leq Score < 4$	88574	131384	152750
$2 \leq Score < 3$	90028	113433	134664
$1 \leq Score < 2$	28316	159991	196379

Table 3: **Distribution of assets and average vertex/face counts for 72B-RR.**

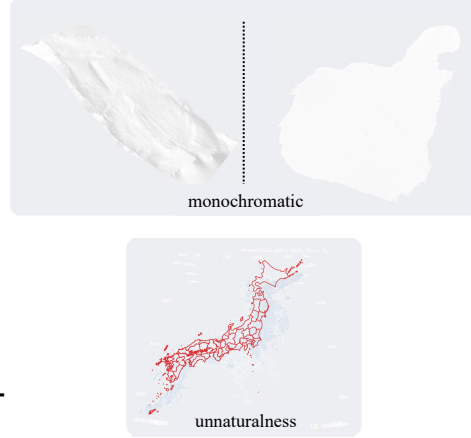


Figure 5: **Examples of textural degradations in Preference-1 examples.** Note that Preference-1 examples include geometrically complex but texturally simple or semantically unrealistic assets, showing that our 72B-RR dataset captures geometric, textural, and semantic quality cues comprehensively.

Dataset	Source	Domain	Annotation	Distortion	Agent	Scalability	Size
WPC [36]	80	Synthetic CAD	MOS (subjective)	Downsampling / Noise / Compression	Human	Low	~4K
LS-PCQA [10]	104	Indoor / Outdoor	Subjective (single)	Voxel / Color / Rendering	Human	Low	~8K
SJTU-PCQA [12]	108	Real + Synthetic	30-user study	Geometry / Texture / Compression	Human	Low-Med	~10K
3D-PAQA (Ours)	260K	Diverse real 3D (Objaverse)	MLLM-aligned (6 criteria)	Natural artifacts	Hybrid	High (auto-scalable)	260K

Table 4: **Comparison of 3D-QA datasets.** Our 3D-PAQA uniquely provides large-scale, MLLM-aligned annotations reflecting natural 3D degradations.

we group the 72B-RR annotated samples into five score intervals ($1 \leq Score < 2, \dots, Score = 5$) and compute the average number of vertices and faces within each group, as shown in Table 3. The results exhibit a clear positive trend: higher-quality objects tend to have larger vertex and face counts, indicating that our 3D-PAQA annotations implicitly capture geometric richness and modeling fidelity. Interestingly, the lowest-quality group ($1 \leq Score < 2$) shows an atypical increase in vertex and face counts. As illustrated in Figure 5, this anomaly arises because low-preference objects often include geometrically dense yet perceptually degraded assets—such as monochromatic or structurally unnatural meshes. This observation suggests that the MLLM-inferred preference reflects not only geometric complexity but also the balance between geometry and texture realism, indicating that 3D-PAQA captures a holistic perception of 3D quality.

Comparison to Prior Datasets. Table 4 compares 3D-PAQA to previous 3D-QA datasets, including WPC [36], LS-PCQA [10], and SJTU-PCQA [12]. Prior datasets are primarily distortion-based and limited to synthetic or hybrid scenes with a few thousand samples, each requiring manual subjective ratings. In contrast, 3D-PAQA scales to over 260K real-world Objaverse assets with automatic MLLM-based annotation, covering six perceptual dimensions. Furthermore, while earlier datasets focus on artificially induced degradation, our dataset captures naturally occurring imperfections such as non-manifold geometry, unrealistic proportions, and material mismatches. This large-scale, human-aligned corpus provides a new foundation for training and evaluating open-domain 3D quality predictors.

4.3 3D-Evaluator

Experimental Details. We train a lightweight point cloud-based evaluator using Point Transformer-v3 PTv3) as the backbone architecture [33]. Each mesh asset is sampled into N points with coordinates and features, including RGB color, surface normals, and PBR attributes (roughness, metallic,

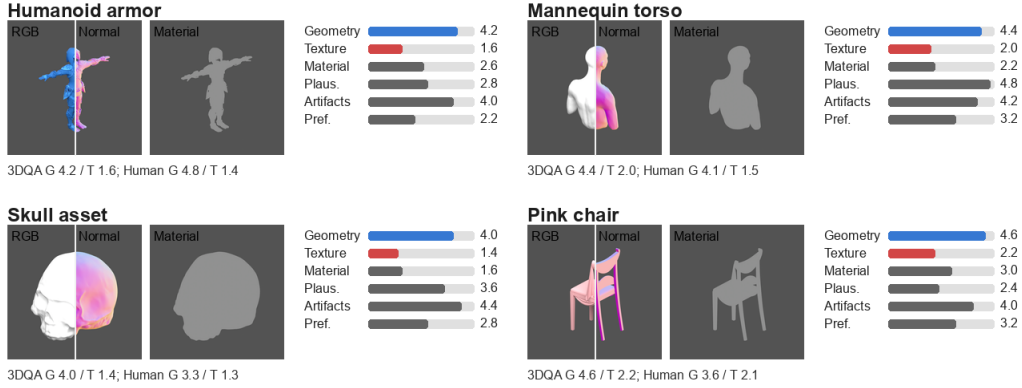


Figure 6: **Diagnosing texture-limited assets with criterion-wise scores.** We visualize assets that receive high geometry scores but low texture scores, indicating that their shapes are coherent while their surface appearance remains weak. RGB renderings are shown alongside normal and material cues to separate geometric quality from appearance quality. These examples are diagnostic targets for texture-focused improvement, not refined outputs.

albedo). When PBR attributes are unavailable, we assign default constant values to maintain feature consistency. These features are processed by the Point Transformer backbone and passed to a two-layer MLP regression head to predict each of 6 criteria. The evaluator is optimized using a Huber regression loss against 3D-PAQA annotations. Training employs AdamW with cosine learning-rate scheduling and standard point cloud augmentations (random rotation, jitter, scaling).

Results. Table 2 presents the quantitative comparison across 72B-RR, Objaverse++, and our trained 3D-Evaluator. The proposed 3D-Evaluator surpasses both baselines in overall correlation with human judgments, achieving the highest PLCC and SROCC across all six perceptual dimensions. Compared to Objaverse++, which relies on limited manual curation, our model demonstrates significantly stronger human alignment with no-manual curation. Notably, the 3D-Evaluator even exceeds its teacher model (72B-RR) on every criterion despite being trained on its annotations. We attribute this to a *distillation effect*, where the model consolidates noisy MLLM supervision into a more stable and human-consistent representation of perceptual quality. These results highlight the potential of the 3D-Evaluator as a lightweight, scalable curator for large-scale 3D datasets.

4.4 Qualitative Results

Figure 7 presents qualitative examples from the 72B-RR annotations across the six perceptual criteria. For each criterion, we visualize representative samples from five rank groups and compare the predicted scores with human judgments. The examples show that the proposed annotation pipeline produces scores that are largely consistent with human perception across a wide quality range.

For component-level criteria (i.e., geometry, texture, and material map), the rank groups reflect visible changes in geometry completeness, texture fidelity, and material consistency. Higher-ranked assets generally contain coherent shapes, detailed surface appearance, and plausible material cues, whereas lower-ranked assets often exhibit missing structures, over-smoothed geometry, weak textures, or mismatched material properties. Since 3D-PAQA decomposes quality into different component-level axes, the resulting scores can also provide actionable guidance for asset refinement. For example, an asset with a high geometry score but a low texture score could be improved by selectively enhancing texture fidelity while preserving its overall shape.

For holistic criteria (i.e., plausibility, artifacts, and overall preference), the annotations capture combined perceptual factors rather than a single visual defect. Assets with unnatural shapes, severe rendering artifacts, or poor semantic consistency receive lower scores even when some local details remain complex. We also observe that most assets receive reasonably high artifact scores, as severe rendering artifacts are relatively uncommon in the sampled set. In contrast, assets with consistently

	Rank group 1		Rank group 2		Rank group 3		Rank group 4		Rank group 5	
Geometry	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal
	Ours(72B-RR): 4.4 / Human: 4.4		Ours(72B-RR): 4.0 / Human: 4.1		Ours(72B-RR): 3.4 / Human: 3.4		Ours(72B-RR): 2.4 / Human: 2.4		Ours(72B-RR): 1.4 / Human: 1.6	
Texture	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal
	Ours(72B-RR): 4.6 / Human: 4.6		Ours(72B-RR): 4.4 / Human: 4.4		Ours(72B-RR): 4.2 / Human: 4.2		Ours(72B-RR): 2.2 / Human: 2.1		Ours(72B-RR): 1.0 / Human: 1.0	
Material	RGB	Material	RGB	Material	RGB	Material	RGB	Material	RGB	Material
	Ours(72B-RR): 4.4 / Human: 4.2		Ours(72B-RR): 3.6 / Human: 3.6		Ours(72B-RR): 2.8 / Human: 3.1		Ours(72B-RR): 1.8 / Human: 2.2		Ours(72B-RR): 1.4 / Human: 1.4	
Plausibility	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal
	Ours(72B-RR): 4.2 / Human: 4.2		Ours(72B-RR): 3.6 / Human: 3.6		Ours(72B-RR): 3.0 / Human: 3.2		Ours(72B-RR): 2.8 / Human: 2.8		Ours(72B-RR): 2.0 / Human: 2.0	
Artifacts	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal
	Ours(72B-RR): 4.4 / Human: 4.4		Ours(72B-RR): 4.2 / Human: 4.2		Ours(72B-RR): 4.0 / Human: 4.1		Ours(72B-RR): 3.6 / Human: 3.6		Ours(72B-RR): 3.4 / Human: 3.4	
Preference	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal	RGB	Normal
	Ours(72B-RR): 4.6 / Human: 4.4		Ours(72B-RR): 3.4 / Human: 3.4		Ours(72B-RR): 3.0 / Human: 2.9		Ours(72B-RR): 2.8 / Human: 2.9		Ours(72B-RR): 1.6 / Human: 1.4	

Figure 7: **Qualitative examples of 3D-PAQA annotations across six perceptual criteria.** Each row shows assets sampled from a specific criterion, and each column corresponds to a rank group from high to low quality. The examples compare 72B-RR predictions with human scores, illustrating that our annotations reflect both component-level degradations, such as geometry, texture, and material quality, and holistic degradations, such as plausibility, artifacts, and overall preference.

Reward backend	\$/eval	Latency	Refinement steps/\$1
3D-Evaluator (ours)	$\approx \$0$	< 1 s	unbounded (local)
MLLM judge	$\approx \$0.064$	~ 15 s	~ 16

Table 5: **Cost of the evaluator-as-verifier.** Per-evaluation cost and latency of the two reward backends, and the number of refinement steps affordable within a fixed \$1 budget. The distilled 3D-Evaluator provides an essentially free local reward, enabling many-iteration refinement where an MLLM judge is quickly budget-bound.

high scores across the holistic criteria tend to exhibit strong realism, coherent structure, and visually natural appearance.

These qualitative trends support the quantitative correlations above and suggest that 3D-PAQA provides interpretable supervision for open-domain 3D quality assessment.

Criterion-wise diagnosis for texture-limited assets. The decomposed annotations also make it possible to identify which component of an asset limits its perceived quality. Figure 6 shows examples with high geometry scores but low texture scores. Their normal cues preserve recognizable silhouettes and surface structure, whereas the RGB renderings and material cues reveal weak, missing, or nearly monochromatic appearance. These examples are not outputs of a refinement method; instead, they demonstrate how 3D-PAQA can flag assets where texture-focused improvement would be more appropriate than geometry modification.

4.5 Evaluator-Guided Asset Refinement

Beyond offline benchmarking, we show that the 3D-Evaluator can act directly as a *verifier inside a generation loop*, turning quality-guided generation into a concrete pipeline. We couple the evaluator with a large language model (LLM) agent that authors procedural content: the agent writes a Blender (bpy) script that builds and exports a mesh, the 3D-Evaluator scores the exported asset on the six perceptual criteria, and the criterion-wise scores together with the current weakest criterion are returned to the agent, which then revises its script to raise that criterion and re-exports. The evaluator thus serves as an efficient, human-aligned *reward* that steers an autonomous generator, replacing repeated MLLM queries or human studies. Because the evaluator is trained on human-created (non-generated) assets, the loop exploits the evaluator’s *decomposed feedback*—which component-level or holistic criterion is weakest, and how the scores move across iterations—rather than the absolute score values.

This closed loop produces visible, targeted improvements. Guided only by the per-criterion feedback, the agent repairs precisely the dimensions the evaluator flags: Figure 8 shows several objects refined left to right, each maturing from a rough initial build into a cleaner, more detailed asset. For the castle, the six 3D-Evaluator scores rise steadily as a garishly coloured, uneven keep becomes a coherent stone castle—texture improves by +1.03, material by +1.02, and overall preference by +1.03 (3.28 $\rightarrow$ 4.32)—with the largest gains on the initially weakest criteria. Because the loop is guided by the same evaluator that reports these scores, we treat the visible, human-verifiable transformation in Figure 8 as the primary evidence, and the rising scores only as corroboration that the evaluator’s signal tracks the qualitative improvement it induces. We report this as a *feasibility* study: the point is that the evaluator supplies an actionable, decomposed signal an agent can optimize against, not that the loop is optimal or that the gains are uniform across objects.

Crucially, using the 3D-Evaluator as the reward makes this loop *economical*. Each evaluation is a single local point-cloud forward pass of the 33M model, costing $\approx \$0$ and completing in under a second, whereas an equivalent MLLM judge costs $\approx \$0.064$ per query and is well over an order of magnitude slower (Table 5). At a fixed budget the cheap, decomposed reward affords far more refinement iterations, which is what makes closed-loop quality-guided generation practical—the same efficiency argument that motivates distilling MLLM supervision into a lightweight evaluator (Section 4.3), now exploited inside the generation loop itself.

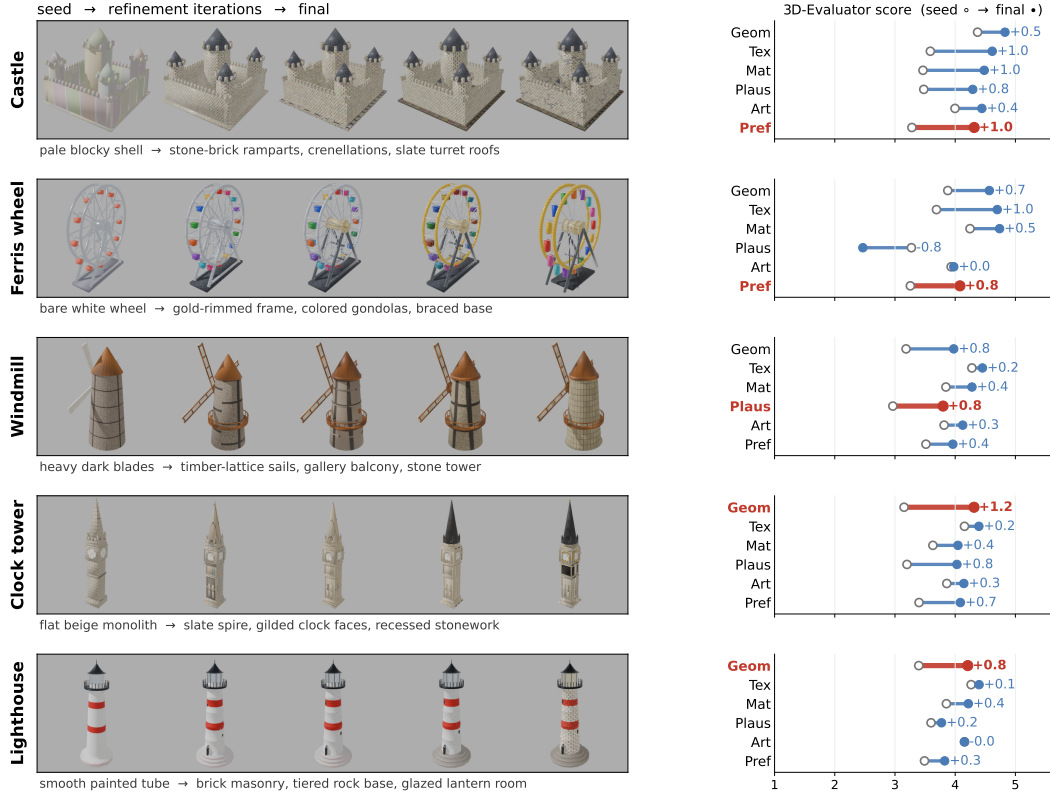


Figure 8: **Evaluator-guided asset refinement.** An LLM agent generates each 3D asset in Blender and iteratively revises it using our 3D-Evaluator as the verifier. *Left:* rendered asset across refinement iterations (seed → final). *Right:* the six 3D-Evaluator scores for each object, from seed (○) to final (●); the criterion that is weakest at the seed is highlighted in red. Guided only by the evaluator’s criterion-wise feedback, each asset matures from a rough initial build into a coherent, detailed one, and the initially weakest criterion is substantially raised.

5 Future Work

Our 3D-PAQA dataset demonstrates strong human alignment, and the trained 3D-Evaluator not only inherits this property but also surpasses its teacher model (72B-RR) in correlation with human judgments. In this section, we discuss further directions suggested by these results. Aside from the evaluator-guided asset refinement already demonstrated in Section 4.5, these directions are not experimentally validated in this work, but outline how the proposed dataset and evaluator could be extended for practical and scalable 3D quality assessment.

5.1 Generative model Evaluator

Our 3D-Evaluator could potentially serve as a metric for assessing the perceptual quality of 3D generative models. Prior approaches have explored this concept through MLLM-based pairwise A/B testing or relative comparison schemes, which require multiple competing models and numerous iterations to estimate consistent rankings. However, these methods cannot assign absolute scores to a single generative model and incur substantial computational overhead due to the use of large vision-language models. In contrast, our lightweight 3D-Evaluator suggests a possible route toward absolute quality prediction with only 33M parameters, providing an efficient and model-agnostic way to evaluate generative outputs. This use case would require additional validation on outputs from modern 3D generative models.

Extensions. Nevertheless, to generalize beyond human-created assets, future work should extend the 3D-PAQA dataset to include annotations on AI-generated artifacts. Incorporating such generative

content may mitigate domain shift and help determine whether the evaluator can consistently assess both human- and AI-generated 3D data.

5.2 Dataset Curator

Our 3D-PAQA (72B-RR) annotations could also support dataset curation for training or evaluating 3D generative models. Since they provide preference-aligned quality scores across diverse object categories, the 72B-RR subset may be useful as a benchmark for data filtering or generative training. Beyond this, our 3D-Evaluator offers a lightweight candidate for scalable assessment. With only 33M parameters, it may reduce the need for costly MLLM inference when filtering large-scale repositories, although its effectiveness as a curator should be verified in downstream training experiments.

5.3 Quality-Guided 3D Generation

Section 4.5 already demonstrates a first realization of this idea: an LLM agent refines generated assets using the 3D-Evaluator’s decomposed, criterion-wise feedback as a reward, without repeated MLLM queries or human studies. That study treats the evaluator as a black-box verifier steering an agent in the loop. A natural next step is tighter integration of the evaluator into the generator itself—for example, through differentiable rendering pipelines, score-distillation objectives, or reinforcement-style optimization that backpropagate the evaluator’s criterion-wise signal directly into 3D generative models—and a systematic study of whether such rewards or losses improve generated assets at scale.

6 Conclusion

We presented 3D-PAQA, a large-scale preference-aligned 3D quality assessment dataset, together with a lightweight 3D-Evaluator trained on its annotations. Our central goal was to move beyond distortion-based 3D-QA settings and establish a more realistic benchmark that reflects how humans actually judge the quality of diverse 3D assets. By leveraging exemplar-anchored prompting and relative ranking with MLLMs, 3D-PAQA captures human perceptual preferences at scale and narrows the gap between subjective human evaluation and automatic 3D quality estimation.

Our experiments show that this dataset is not only large and diverse, but also practically useful as a source of supervision for downstream quality prediction. In particular, the resulting 3D-Evaluator achieves strong correlation with human judgments across all six perceptual dimensions, while remaining far more efficient than large multimodal models. Notably, the evaluator even surpasses its teacher MLLM on every criterion, suggesting that noisy but large-scale multimodal supervision can be distilled into a compact and stable predictor. Qualitative analyses further show that the proposed criteria provide interpretable feedback: component-level scores reveal specific weaknesses in geometry, texture, and material quality, while holistic scores capture realism, artifacts, plausibility, and overall preference.

Beyond benchmarking, our framework suggests several directions for practical 3D quality control. The proposed evaluator may be useful as an efficient metric for generative model assessment or as a lightweight dataset curator for large-scale 3D repositories, potentially reducing the reliance on repeated pairwise comparisons, costly human studies, or heuristic filtering rules. Its decomposed quality predictions also provide actionable guidance for asset refinement: we demonstrate this in Section 4.5, where the evaluator acts as an efficient reward in an agentic loop that steers an LLM agent to raise the weakest criterion of a generated asset over successive iterations. More broadly, future work could integrate the evaluator more tightly into quality-guided 3D generation pipelines, where its criterion-wise scores act as differentiable rewards or losses for improving generated assets.

Looking ahead, extending 3D-PAQA to include more AI-generated assets may improve generalization and help evaluate artifacts specific to modern generative pipelines. We hope that 3D-PAQA will serve as a foundation for future research on unified, human-aligned, and scalable 3D quality assessment across both human-created and machine-generated content.

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